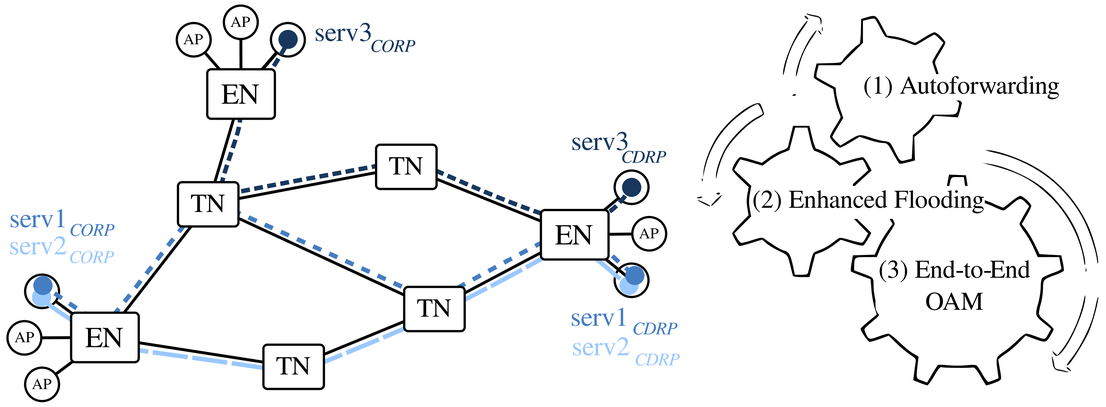
TOWARDS A LARGE SCALE LOCARN DESIGN

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WHAT IS LOCARN ?

" LOW OPEX CAPEX ARCHITECTURE FOR RESILIENT NETWORKS "



EN Edge Node TN Transit Node
AP Access Port Service *Origin/Destination*

(1) Each data packet contains in header the information required to be transported from end-to-end across the network. The forwarding process *does not involve any state* in the data plane of Transit Nodes (neither transmission table nor lookup process).
(2) On services registration the network is flooded to *discover* possible paths; extra information is retrieved to characterize each of them. The best path is selected at edges by simple comparison.
(3) Operation And Maintenance: the service continuity is checked by continuous exchanges of small packets from end-to-end.

dramatically simple packet layer (↘ CAPEX) ; very adaptive and opportunistic (↘ OPEX); tunable resiliency (even <50ms service recovery)

A SCALABILITY PROBLEM

Problem: The amount of packets produced by LOCARN (overhead) **linearly depends on the number of services** because each one involves its own routing and fault detection process. The design is said "service oriented" (studied in [1]).

Solution: To increase the architecture scalability, we want to **decorelate the overheads produced from the number of services**
⇒ we define an improved *"Large Scale Design"*:

- **Proposal 1 = Control Plane decorrelation**
- **Proposal 2 = OAM decorrelation**.

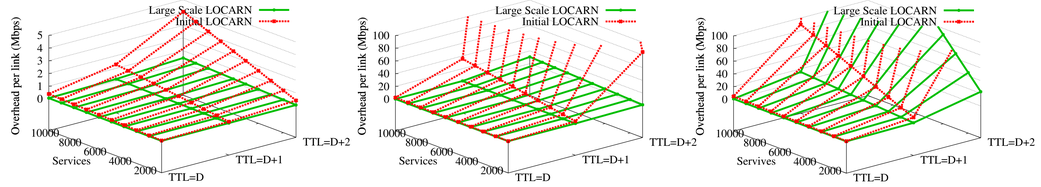
SIMULATION TOOLS

LOCARN simulation *OMNeT++* (C++)
N-ary trees structures *STLplus* library (C++)
Networks topologies *CCNSim* simulator (C++)

Network	Nodes	Links	Density	Diameter
GEANT	22	37	3.4	4
Dtelecom	68	353	10.4	3
Level 3	46	268	11.7	4

Table 1: Networks Topologies Dimensions

PROPOSAL 1: MULTI-SERVICES PATH DISCOVERY



GEANT Network DTELECOM Network LEVEL 3 Network

Figure 1: Initial design beside multi-services path discovery (realistic topologies)

- **Idea** No need to send several floodings to discover services paths from a same node
- **Implementation** Send multi-services path discoveries = flooding packets include several services identifiers rather than a single one
- **Consequences** Flooding packets size ↗ *a little* but the amount of floodings ↘ *a lot*
- **Results** The LOCARN control plane is drastically more scalable in terms of services amount

PROPOSAL 2: POINT-TO-MULTIPOINT AUTOFORWARDING

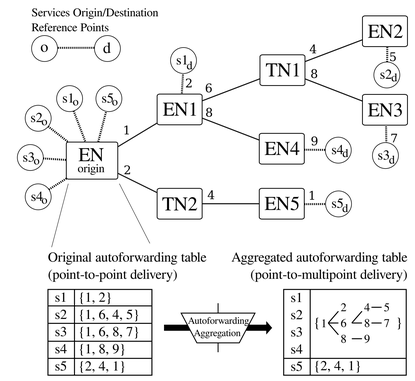
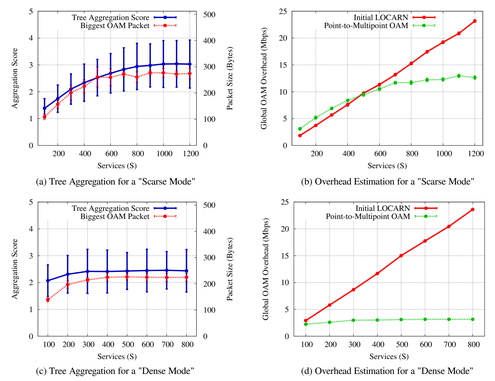


Figure 2: Illustration of P2MP autoforwarding



(a) Tree Aggregation for a "Scarse Mode" (b) Overhead Estimation for a "Scarse Mode"
(c) Tree Aggregation for a "Dense Mode" (d) Overhead Estimation for a "Dense Mode"

Figure 3: P2MP Simulation Results (GEANT Network)

- **Idea** When each service checks its active path, some links are checked several time unnecessarily
- **Implementation** Point-to-multipoint (P2MP) autoforwarding: (i) at the source, a P2MP packet header contains a "flattened n-ary tree structure" (ii) packets are forwarded and/or duplicated until destinations.
- **Consequences** OAM packets size tend to ↗ whereas their amount tends to ↘
- **Results** The OAM overhead is clearly decorrelated from the services amount, both in a scarse and in a dense distribution of endpoints

REFERENCES

[1] D. Le Quere, C. Betoule, R. Clavier, Y. Hadjadj-Aoul, A. Ksentini, and G Thouenon. Scalability & Performances Evaluation of LOCARN: Low Opex and Capex Architecture for Resilient Networks. In *I4CS 2014. 14th International Conference on Innovations for Community Services*, June 2014.

CONCLUSIONS

- **LOCARN is now able to scale up to 10000 services** without impact over reasonable backbone topologies
- **The two proposals involve (few) more complexity in Edge and Transit nodes**
- **Used for data packets forwarding, the proposal 2 allows LOCARN to achieve "stateless multicast"** without involving any additional protocol

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